Dual-ligand screen / Setup

48 pairs, then 22 pairs + L17 reference. One 48 x 2 mL Para-Dox, reused with **71 fresh reaction vials**. Manual screw caps; no stir bars. Keep reaction and LC maps separate.

Before the clock

- Fit **p300 single GEN2 left; p300 multi GEN2 right**. Single only for 13 mm-pitch reaction vials. Check block/latch, offsets, cap height <=45 mm and tips >1 mm above bottoms. Simulate released Python locally.
- Use matching `dosing.csv`, `stock_preparation.csv`, `deck_loads.csv`, `lc_sequence.csv` and `round2_predose.csv`. A software colleague handles release and LC imports. Stocks and initial scoring are outside the clock.
- Have 71 reaction vials/caps, 71 autosampler-compatible HPLC vials/caps plus standards, two 96-well deep-well preparation plates matching the supplied VWR 2.2 mL definition, fresh tip racks and >=71 manual sample tips. Confirm the **full 1.7-min LC cycle** and batch handoff.

Prepare stocks

Stock	Bulk preparation / use
SM in DCM	**500 mM, 10 mL:** 805.8 mg SM at 100% assay; capped 20 mL glass vial. Dose 50 µL; evaporate fully.
Ligands / Pd	**40 mM in toluene.** Use CSV masses/reserves. Open and prepare ligands in glovebox. Keep reserves capped.
Mesitylene	**100 mM in toluene, 5 mL:** 60.095 mg. Dose 25 µL AFTER DCM removal.
Quench	**100 mL, 9:1 ACN/water +0.15% formic acid:** 150 µL acid, make to volume. No IS.
LC diluent	**125 mL, 1:1 water/ACN +0.1% formic acid:** 125 µL acid, make to volume. No IS.

Correct masses for assay. **Load stage-specific CSV volumes, not bulk preparations.** Refill R2 from reserves; ordinary plastic source racks are sufficient.

Deck

Slot	Contents
1 / 4	Ligand source racks; exact CSV addresses; L07 absent
2	A1 SM/DCM; A2 Pd; A3 toluene; A8 mesitylene. Quench A4/A6/A9; diluent A5/A7/A10/A11/A12
3 / 10	Reaction block: staff move it by hand at App prompts; pipetting / heater-shaker
6	96-well 2.2 mL deep-well LC preparation plate; stays here, no module shaking
5 / 8 / 9	Single tips / multi tips / multi tips. Keep 7 and 11 empty

Calibration during hold one - 14 injections

Each SM and product: blank + **10, 50, 150, 350, 800 µM**, plus independent **666.7 µM** check. **Mesitylene stays at 66.7 µM in every vial.** Use matched matrix and calibration recipes. HTEL exports UV areas; Python calibrates and calculates yields. The software colleague handles method/RT settings and pilot checks.

Dual-ligand screen / Run

Operator / start: _____ LC method / channel: _____

Follow App pauses. Open only the indicated vials; hand-tighten all caps before heating or shaking. Record actual times and deviations.

Step	Action
1 / SM	Dose **50 µL SM/DCM** into 48 vials. At pause, move off deck; evaporate fully by HTEL's ventilated procedure. Cool; return to slot 3.
2 / Reaction	Add **25 µL mesitylene +37.5 µL ligand A +37.5 µL ligand B +75 µL Pd +75 µL toluene =250 µL**. Hand-cap; move/latch as prompted. **95 °C, 500 rpm, 4 h.**
3 / During hold	Run calibration. Hand-dose **23 labelled NEW vials with 50 µL SM/DCM each** in an off-deck rack. Evaporate fully; cap dry SM. No mesitylene/catalyst yet.
4 / Cool	Move the sealed block off the heater. Cool before opening. Load workup and its matching sources.
5 / Quench	Add **500 µL quench** per vial. Hand-recap ALL. Shake **1500 rpm, 2 min**, heater off; return to slot 3.
6 / Sample	Remix the indicated capped batch; let bubbles settle, then uncap only those vials. Robot adds **20 µL extract to 980 µL diluent** with fresh sample tips.
7 / LC handoff	At pause, recap reaction vials. LC plate stays in slot 6; no shaker move. Fresh manual tip per completed 1 mL well: **mix 500 µL x5**, avoiding air; transfer **900 µL** to labelled HPLC vial. Cap/invert. **Start LC batch before resuming robot.** Never take unfinished wells.
8 / Select R2	After all **48 results** pass QC, software selects **22 new pairs + L17 reference**. Archive R1 vials; fit the 23 dry predosed vials into the SAME block. Check orientation.
9 / R2	Refill from R2 deck loads. **No SM redose.** Add selected stocks; reference uses **75 µL L17**, no ligand B. Cap; repeat 4 h reaction/workup. Sample 8/8/7. LC well **H3 is diluent only: do not inject.**

Stop the clock after **23 final injections and quantitative analysis**. Save exports, maps, QC, protocols, times and robot JSON logs.

If a transfer looks wrong

Pause for layers, solids at aspiration, persistent bubbles, air uptake or tip contact. Repeated aspiration does not fix solubility. Reuse tips only for approved same-stock/noncontact dosing; samples always get fresh tips. Replace an uncertain dilution with a fresh documented one; do not guess a top-up.

Timing: 717.7 /720 min planned, including 35 min LC/handling overlap; unmeasured, with about two minutes spare. Full procedure and local commands.